Week 14 — Extended Reading: Offline reinforcement learning

STUDY-RL

Source audit: 20 October 2026

Read for a claim, an assumption and a test

Read the teaching guide first, then use this pack alongside the lecture. The annotations below are course-written paraphrases and investigation prompts, not quotations from the papers. Each source link identifies a primary paper or official guidance. Benchmark results do not transfer automatically to our small exercises. For each reading, write the mathematical object, the needed assumptions and one observation that would challenge your interpretation.

1 Levine et al. (2020): Offline RL tutorial

Primary source. https://arxiv.org/abs/2005.01643

Mechanism and scope. The survey organises the offline problem around distribution shift and limited feedback. A fixed log prevents the learner from correcting arbitrary errors by gathering new transitions. Read the policy-constraint, conservative-value and evaluation discussions as different responses to that problem.

Read, derive, investigate. Draw the dataset distribution and the distribution induced by an improved policy. Which expectation appears in the Bellman loss, and which one matters at deployment? Construct a missing-action example in which an arbitrarily large critic prediction is consistent with the observed data.

2 Kumar et al. (2020): Conservative Q-Learning

Primary source. https://arxiv.org/abs/2006.04779

Mechanism and scope. CQL adds a conservative critic regulariser to Bellman fitting. The discrete log-sum-exp term pushes down actions relative to dataset actions. Lower-bound results have statistical and optimisation conditions; a trained neural critic is not a universal pointwise lower bound.

Read, derive, investigate. At a state with four zero-valued actions and only action 0 in the dataset, differentiate $\log \sum_a e^{Q_a} - Q_0$. Why is its gradient $(-3/4, 1/4, 1/4, 1/4)$? Explain what a large penalty can sacrifice when the data omit useful behaviour.

3 Kostrikov et al. (2021): Implicit Q-Learning

Primary source. https://arxiv.org/abs/2110.06169

Mechanism and scope. IQL uses asymmetric squared-error regression for the state value, then Bellman fits and advantage-weighted policy extraction. Its value update avoids maximising over arbitrary newly sampled actions. Neural policy extraction can still generalise beyond observed actions.

Read, derive, investigate. For equal-probability targets 0 and 1, solve the expectile stationarity equation and obtain $v = \tau$. Why is this an expectile rather than a quantile? Trace which network is held fixed at each update; a high expectile is not an uncertainty estimate.

4 Fujimoto and Gu (2021): TD3+BC

Primary source. https://arxiv.org/abs/2106.06860

Mechanism and scope. TD3+BC combines an actor value objective with behaviour cloning and state normalisation. The published implementation uses $\lambda = \alpha / \operatorname{mean} |Q|$ in the loss $-\lambda Q + \mathrm{BC}$, with the scale detached.

Read, derive, investigate. Multiply critic values by 10 and recompute λ. Which part of the actor objective is approximately scale-invariant? Compare against incorrectly multiplying the BC term by the inverse-value scale. Locate the exact actor loss in the author code before changing the convention.

5 Tarasov et al. (2023): ReBRAC

Primary source. https://arxiv.org/abs/2305.09836

Mechanism and scope. ReBRAC revisits a regularised actor–critic with design changes including deeper networks, LayerNorm, larger batches and decoupled penalties. Its ablations study particular datasets and evaluation protocols; the paper is not a timeless ranking of offline algorithms.

Read, derive, investigate. Create an ablation table with one changed design choice per row. Which comparisons require equal environment data, tuning budget and seeds? The workbook illustrates losses and extraction; it does not reproduce the complete ReBRAC benchmark.

6 Voloshin et al.: Empirical OPE study

Primary source. https://arxiv.org/abs/1911.06854

Mechanism and scope. The study compares off-policy estimators across experimental settings. Fitted Q Evaluation performs Bellman regression for a fixed policy without learning a transition model. Importance weighting instead relies directly on support and correctly aligned probabilities.

Read, derive, investigate. Separate finite-sample bias, variance and model error. A deterministic target does not itself invalidate OPE; missing behaviour support does. The course log experiment is an estimator diagnostic, not evidence that one method dominates in every task.

Connect the reading to your workbook

Use the workbook's expectile and CQL exercises first, then perturb the dataset coverage while keeping the evaluation task fixed. Report the estimate, support diagnostic and uncertainty together. For policy extraction, distinguish the exact nonparametric KL-constrained solution from its weighted maximum-likelihood projection into a neural family.

Evidence boundary. Source identities and the central claims above were checked against primary material. These annotations do not reproduce every proof, training run or empirical result. Use the original paper for theorem assumptions and experimental details; use `project/` and the workbook for the exact course implementation.